N23 Representation of atomic structure

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Reference: Atom-density representations for machine learning, M. J. Willatt, F. Musil, M. Ceriotti, Journal of Chemical Physics, 2019.

We wish to find a function f that takes in a structure, and spits out a property of interest y, let us assume y is energy for the rest of this note for simplicity. Given a data set, machine learning finds this function via curve fitting by adjusting the free parameters of this function f, although typically in high dimensions. Structure is typically regarded a set of positions of atoms and their chemical identity, however, the structure is a very physical thing and contains the geometric relationships between atoms, and it is independent of the typical Cartesian representation of it, or any coordinate system. We associate ket $|A\rangle$ that describes the elemental composition and geometric arrangement of atoms, it contains complete information of the structure (ket is just a high dimensional vector, vector is just a line segment with a direction, which is clearly independent of coordinates) we let ket $|\alpha\rangle$ denote the elemental identity, it satisfy $\langle\alpha|\alpha'\rangle = \delta_{\alpha\alpha'}$ since an atom cannot be two elements at the same time. To make this concrete, project $|A\rangle$ into position representation

$$\langle \mathbf{r}|A\rangle = \sum_i g(\mathbf{r}-\mathbf{r}_i)|\alpha_i\rangle \tag{1.1}$$

where i is an atom index and the sum is taken over all atoms in the structure, $\mathbf{r}_i$ is the position of atom i. $g(\mathbf{r}-\mathbf{r}_i)$ is a Gaussian centred at position of atom i, $\mathbf{r}_i$. Integrating over the space occupied by the structure gives

$$\int d\mathbf{r}\,\langle \mathbf{r}|A\rangle = \sum_i |\alpha_i\rangle = \sum_\alpha N_\alpha|\alpha\rangle \tag{1.2}$$

which counts the number of atoms for each elements. If there is are only one element, for example 1000 carbon atom in a diamond, the integral is $1000|\mathrm{C}\rangle$. The function we want to find is $y = f(|A\rangle)$, f could be a long list of linear and non-linear operations on $|A\rangle$ followed by a final projection onto some bra $\langle\cdot|$. Symmetry operations $\hat{S}$ such as translation, rotation of the whole structure and permutation of two of the same type of atoms should not change the energy of the structure. In other words, we should have

$$f(|A\rangle) = f(\hat{S}|A\rangle) \tag{1.3}$$

suppose there are only two operations: do nothing $\hat{1}$, and inversion $\hat{P}$, with $\hat{P}^2 = \hat{1}$, then define

$$|A^{(1)}\rangle_{\hat{P}} \equiv \frac{1}{2}\Big(|A\rangle + \hat{P}|A\rangle\Big) \tag{1.4}$$

the reason for the superscript (1) will become apparent later, then

$$\hat{P}|A^{(1)}\rangle_{\hat{P}} = \frac{1}{2}\left(\hat{P}|A\rangle + \hat{P}^2|A\rangle\right) = |A^{(1)}\rangle_{\hat{P}} \tag{1.5}$$

so $|A^{(1)}\rangle_{\hat{P}}$ by construction is invariance under $\hat{P}$. So if one replace $|A\rangle \to |A^{(1)}\rangle_{\hat{P}}$ in Eq.(1.3), it is immediately satisfied. Now consider a rotations around an axis $\hat{R}(\theta)$, then the symmetry averaged ket is

$$|A^{(1)}\rangle_{\hat{R}} \equiv \frac{1}{2\pi}\int_0^{2\pi} d\theta\, \hat{R}(\theta)|A\rangle \tag{1.6}$$

if $|A\rangle$ is already rotationally averaged

$$\hat{R}(\theta)|A\rangle = |A\rangle \tag{1.7}$$

so

$$|A^{(1)}\rangle_{\hat{R}} = \frac{1}{2\pi}\int_0^{2\pi} d\theta\, \hat{R}(\theta)|A\rangle = \frac{1}{2\pi}\int_0^{2\pi} d\theta\, |A\rangle = |A\rangle \tag{1.8}$$

therefore the factor of $1/2\pi$ is required for consistency. We show $|A^{(1)}\rangle_{\hat{R}}$ is invariant to rotation by rotating the new rotationally averaged ket by angle α

$$\hat{R}(\alpha)|A^{(1)}\rangle_{\hat{R}} = \frac{1}{2\pi}\int_0^{2\pi} d\theta\, \hat{R}(\alpha)\hat{R}(\theta)|A\rangle = \frac{1}{2\pi}\int_0^{2\pi} d\theta\, \hat{R}(\theta+\alpha)|A\rangle \tag{1.9}$$

let $\theta' = \theta + \alpha$, so $d\theta' = d\theta$, and use the fact that $\int_0^{2\pi} = \int_\alpha^{\alpha+2\pi}$, we have

$$\hat{R}(\alpha)|A^{(1)}\rangle_{\hat{R}} = |A^{(1)}\rangle_{\hat{R}} \tag{1.10}$$

so $|A^{(1)}\rangle_{\hat{R}}$ is indeed rotationally invariant. Now consider rotation in 3D, $\hat{R}(\boldsymbol{\theta})$ where $\hat{\boldsymbol{\theta}} = \boldsymbol{\theta}/\theta$ is the unit vector of rotation axis and $\theta = |\boldsymbol{\theta}|$ is the anti-clockwise angle about that axis. We restrict $0 \le \theta \le \pi$ because $\pi < \theta < 2\pi$ can be described by a rotation with angle $< \pi$ with a rotation axis in the opposite direction $-\hat{\boldsymbol{\theta}}$, the rotationally averaged ket is

$$|A^{(1)}\rangle_{\hat{R}} = \frac{1}{2\pi^2}\int d^3\boldsymbol{\theta}\, \frac{\sin^2(\theta/2)}{\theta^2}\hat{R}(\boldsymbol{\theta})|A\rangle \tag{1.11}$$

which can be more succinctly written as

$$|A^{(1)}\rangle_{\hat{R}} = \int d\hat{R}\, \hat{R}|A\rangle \tag{1.12}$$

with

$$d\hat{R} = \frac{1}{2\pi^2}\frac{\sin^2(\theta/2)}{\theta^2}d^3\boldsymbol{\theta} \tag{1.13}$$

see appendix A for where the $\sin^2(\theta/2)/\theta^2$ comes from. For transnationally averaged ket with translation operator $\hat{T}(\mathbf{t})$ moves all atoms by the same vector $\mathbf{r}_i \to \mathbf{r}_i + \mathbf{t}$

$$|A^{(1)}\rangle_{\hat{T}} = \int d^3\mathbf{t}\, \hat{T}(\mathbf{t})|A\rangle \tag{1.14}$$

where the integral is taken over all 3D space, so for translation $d\hat{T} = d^3\mathbf{t}$.

$$\begin{aligned}\hat{T}(\mathbf{a})|A^{(1)}\rangle_{\hat{T}} &= \int d^3\mathbf{t}\hat{T}(\mathbf{a}+\mathbf{t})|A\rangle \\ &= \int d^3\mathbf{t}'\hat{T}(\mathbf{t}')|A\rangle \\ &= |A^{(1)}\rangle_{\hat{T}}\end{aligned} \tag{1.15}$$

so the transnationally averaged ket is invariant under translation. In general a symmetry averged ket is defined as

$$|A^{(1)}\rangle_{\hat{S}} \equiv \int d\hat{S}\,\hat{S}|A\rangle \tag{1.16}$$

where $d\hat{S}$ is chosen so that $\hat{S}|A^{(1)}\rangle_{\hat{S}} = |A^{(1)}\rangle_{\hat{S}}$ is satisfied. Now consider the position representation of the ket, using the fact that g is a normalised Gaussian

$$\begin{aligned}\langle\mathbf{r}|A^{(1)}\rangle_{\hat{T}} &= \int d^3\mathbf{t}\,\langle\mathbf{r}|\hat{T}(\mathbf{t})|A\rangle \\ &= \int d^3\mathbf{t}\,\langle\mathbf{r}-\mathbf{t}|A\rangle \\ &= \sum_i |\alpha_i\rangle \int d\mathbf{t}\,g(\mathbf{r}-\mathbf{t}-\mathbf{r}_i) \\ &= \sum_\alpha N_\alpha|\alpha\rangle\end{aligned} \tag{1.17}$$

which is clearly not good, because this is now just a sum over number of atom of each element and we have loss information on the geometric structure. To avoid this issue, define

$$|A^{(\nu)}\rangle_{\hat{T}} \equiv \int d^3\mathbf{t}\,(\hat{T}(\mathbf{t})|A\rangle)^{\otimes\nu} \tag{1.18}$$

so that $\hat{T}^{\otimes\nu}|A^{(\nu)}\rangle_{\hat{T}} = |A^{(\nu)}\rangle_{\hat{T}}$ where tensor power is defined as

$$|\psi\rangle^{\otimes\nu} \equiv \underbrace{|\psi\rangle\otimes|\psi\rangle\cdots|\psi\rangle}_{\nu \text{ times}} \tag{1.19}$$

notice this is consistent with notation above, which is the case of $\nu = 1$. For $\nu = 2$ we have

$$\langle\mathbf{r},\mathbf{r}'|A^{(2)}\rangle_{\hat{T}} = \sum_{ij}|\alpha_i\alpha_j\rangle\int d\mathbf{t}\,g(\mathbf{r}-\mathbf{t}-\mathbf{r}_i)g(\mathbf{r}'-\mathbf{t}-\mathbf{r}_j) \tag{1.20}$$

define cross correlation $\star$ between two functions via

$$(f\star g)(t) = \int dt'\,f(t')g(t'+t) \tag{1.21}$$

then use substitution $\mathbf{u} = \mathbf{r}'-\mathbf{t}-\mathbf{r}_j$, define $\Delta\mathbf{r} = \mathbf{r}-\mathbf{r}'$ and $\mathbf{r}_{ij}$, we have

$$\begin{aligned}\langle\mathbf{r},\mathbf{r}'|A^{(2)}\rangle_{\hat{T}} &= \sum_{ij}|\alpha_i\alpha_j\rangle\int d^3\mathbf{u}\,g(\mathbf{u})g(\mathbf{u}+\Delta\mathbf{r}-\mathbf{r}_{ij}) \\ \langle\Delta\mathbf{r}|A^{(2)}\rangle_{\hat{T}} &= \sum_{ij}|\alpha_i\alpha_j\rangle(g\star g)(\Delta\mathbf{r}-\mathbf{r}_{ij})\end{aligned} \tag{1.22}$$

which retains the geometric features of the structure. Now consider energy y as a linear prediction, given by the projection

$$y(A) = \langle y|A^{(2)}\rangle_{\hat{T}} \tag{1.23}$$

relabel $\Delta\mathbf{r} \to \mathbf{r}$ then insert a complete set of states gives

$$y(A) = \int d^3\mathbf{r}\, y(\mathbf{r})\langle \mathbf{r}|A^{(2)}\rangle_{\hat{T}} \tag{1.24}$$

where we have define real function $y(\mathbf{r}) = \langle \mathbf{r}|y\rangle = \langle y|\mathbf{r}\rangle$, which is to be learnt by the model in order to make prediction. Now, consider the structure consists of only one chemical species, and take the Dirac delta limit of the Gaussian

$$(g \star g)(\mathbf{r} - \mathbf{r}_{ij}) \to \int d^3\mathbf{u}\, \delta(\mathbf{u})\delta(\mathbf{u} + \mathbf{r} - \mathbf{r}_{ij}) = \delta(\mathbf{r} - \mathbf{r}_{ij}) \tag{1.25}$$

and we have

$$\langle \mathbf{r}|A^{(2)}\rangle_{\hat{T}} = \sum_{ij} \delta(\mathbf{r} - \mathbf{r}_{ij}) \tag{1.26}$$

then after angular averaging, then average over ensemble, this is just a histogram of the distances between atoms, in other words, the radial distribution function (RDF). Substitute into $y(A)$ gives

$$y(A) = \sum_{ij} \int d^3\mathbf{r}\, y(\mathbf{r})\delta(\mathbf{r} - \mathbf{r}_{ij}) = \sum_{ij} y(\mathbf{r}_{ij}) \tag{1.27}$$

$y(\mathbf{r}_{ij})$ is a pair potential, which has an intimate connection to RDF and $\langle \mathbf{r}|A^{(2)}\rangle_{\hat{T}}$. (See Henderson's theorem for more detail) It is easy to show for g not in Dirac delta limit, we have

$$y(A) = \sum_{ij} [y \star (g \star g)](-\mathbf{r}_{ij}) = \sum_{ij} [y \star g^{\star 2}](-\mathbf{r}_{ij}) \tag{1.28}$$

The overlap $y(A) = \langle y|A^{(2)}\rangle_{\hat{T}}^{\otimes 2}$, with two complete set of position eigenstates inserted is

$$y(A) = \sum_{iji'j'} \int d^3\mathbf{r}\, d^3\mathbf{r}' y(\mathbf{r}, \mathbf{r}')h(\mathbf{r} - \mathbf{r}_{ij})h(\mathbf{r}' - \mathbf{r}_{i'j'}) \tag{1.29}$$

which is

$$y(A) = \sum_{iji'j'} [y \star g^{\star 2}](-\mathbf{r}_{ij}, \mathbf{r}_{i;j'}) \tag{1.30}$$

in the limit of $h \to \delta$ we have

$$y(A) = \sum_{iji'j'} y(\mathbf{r}_{ij}, \mathbf{r}_{i'j'}) \tag{1.31}$$

and similarly for ν tensor powers of $|A^{(2)}\rangle_{\hat{T}}$, there would be ν-pairs of ij in the sum and ν arguments in the function $y \star g^{\star 2}$

Eq.(1.22) suggests the factorisation

$$|A^{(2)}\rangle_{\hat{T}} = \sum_j |\alpha_j\rangle|\rho_j\rangle \tag{1.32}$$

where $|X_i\rangle$ is the environmental atom centred ket, define $h = g \star g$, the positon representation of it is

$$\langle \mathbf{r}|\rho_j\rangle = \sum_i f_c(r_{ij}) h(\mathbf{r} - \mathbf{r}_{ij})|\alpha_i\rangle \tag{1.33}$$

which is the environment of atom i. $f_c(r_{ij})$ is a smooth cutoff, for example $f_c(r) = 1/2 + \cos(\pi r/r_c)/2$, which decays to zero along with all its derivatives at cutoff radius r_c. So the environment of the atom only depends on a spherical neighbourhood centred on it. The notation is further simplified by defining

$$\rho_j^\alpha(\mathbf{r}) \equiv \langle \alpha, \mathbf{r}|\rho_j\rangle = \sum_{i\in\alpha} f_c(r_{ij}) h(\mathbf{r} - \mathbf{r}_{ij}) \tag{1.34}$$

so

$$\langle \mathbf{r}|\rho_j\rangle = \sum_\alpha \rho_j^\alpha(\mathbf{r})|\alpha\rangle \tag{1.35}$$

where we used the elemental kets are complete in the elemental space

$$\hat{1} = \sum_\alpha |\alpha\rangle\langle\alpha| \tag{1.36}$$

in general

$$|A^{(\nu+1)}\rangle_{\hat{S}} = \sum_i |\alpha_i\rangle|\overline{\rho_i^{\otimes\nu}}\rangle \tag{1.37}$$

Appendix A

Define a 4D vector q

$$q \equiv \left(\cos\frac{\theta}{2}, \hat{\boldsymbol{\theta}}\sin\frac{\theta}{2}\right) \tag{A.1}$$

more explicitly

$$q = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2)\sin\alpha\cos\phi \\ \sin(\theta/2)\sin\alpha\sin\phi \\ \sin(\theta/2)\cos\alpha \end{pmatrix} \tag{A.2}$$

where we have parametrised the vector on unit sphere $\hat{\boldsymbol{\theta}}$ by a colatitude α and an azimuth ϕ. It immediately follows from Eq.(A.1) that

$$q^2 = 1 \tag{A.3}$$

so it is a unit vector on a 4D sphere, also known as a unit **quaternion**. For a fixed θ, the last three components of q is a lies on a 3D sphere with radius $\sin(\theta/2)$. A small patch on that sphere has area

$$dA = \sin^2\frac{\theta}{2}d\Omega \tag{A.4}$$

where $d\Omega = \sin\alpha\, d\alpha\, d\phi$ is small solid angle, with integral

$$\int d\Omega = 4\pi \tag{A.5}$$

for fixed rotation axis $\hat{\boldsymbol{\theta}}$, q moves around a unit circle in the plane spanned by the first coordinate the $\hat{\boldsymbol{\theta}}$ (To make this more concrete, let $\hat{\boldsymbol{\theta}}$ be imaginary unit i, then $q = (\cos(\theta/2), i\sin(\theta/2))$ is just a unit circle in complex plane) But notice that when physical angle change by $d\theta$, the angle around this circle only changes by $\theta/2$, using arc length is radius $\times$ angle, the distance moved on circle is $1 \times d\theta/2$, then multiply this by patch area dA gives the small three dimensional volume at angle $(\theta/2, \alpha, \phi)$ on the surface of this 4D unit sphere

$$dV = \frac{1}{2}\sin^2\frac{\theta}{2}d\theta\, d\Omega \tag{A.6}$$

note

$$\int dV = \frac{1}{2}\int d\Omega \int_0^\pi d\theta\, \sin^2\frac{\theta}{2} = \pi^2 \tag{A.7}$$

for $\hat{R}|A\rangle = |A\rangle$, we require $|A^{(1)}\rangle_{\hat{R}} = |A\rangle$, this is same as requiring

$$\int d\hat{R} = 1 \tag{A.8}$$

clearly, this is satisfied by

$$d\hat{R}(\boldsymbol{\theta}) = \frac{dV}{\pi^2} \tag{A.9}$$

so integrating over all rotations in 3D is same as integrating over the 3D volume boundary on a 4D unit sphere, then dividing by $2\pi^2$. (Remember that we only integrated over half

of the boundary here of this 4D sphere, because of our earlier restriction that $0 \leq \theta \leq \pi$, so we only need to divide by π^2 instead of $2\pi^2$) Use

$$d^3\boldsymbol{\theta} = \theta^2 d\Omega \, d\theta \tag{A.10}$$

where $\theta^2 \, d\Omega$ is small patch area at radius θ, and $d\theta$ is a small change in radial distance. We finally have

$$d\hat{R}(\boldsymbol{\theta}) = \frac{1}{2\pi^2} \frac{\sin^2(\theta/2)}{\theta^2} d^3\boldsymbol{\theta} \tag{A.11}$$